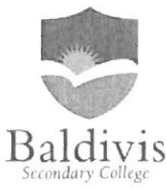
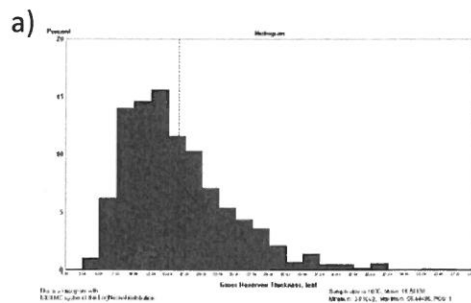


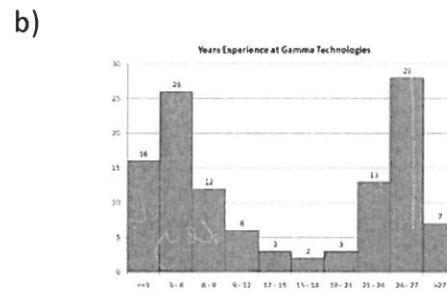
Name:	<u>Answers - Version 1</u>		Date: _____
Teacher:	_____		
	Year 9 Mathematics Mini test 4 Topic – Data Analysis		SCORE: / 28
	<u>Full working out MUST be shown to get full marks for each question.</u>		
Total Time:	30 minutes		
Weighting:	5 %		
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.		

1) Underneath each graph below, write the most appropriate term to describe the distribution.

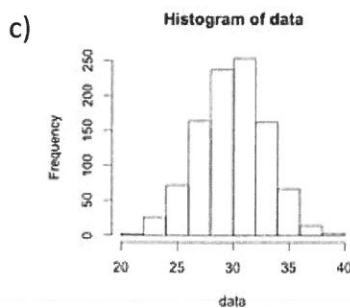
[4 marks]



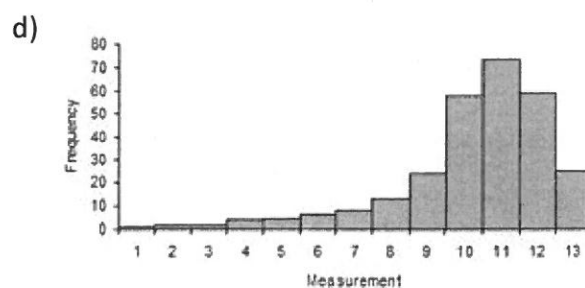
Positive skew



bimodal



Symmetrical



negative skew

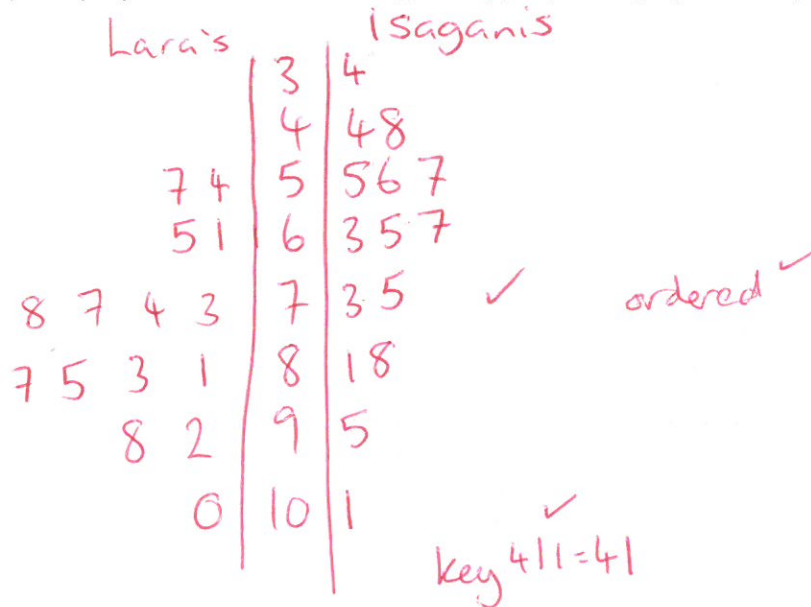
2. Isagani and Lara were arguing whose basketball team was better. They looked at the scores from their games last year: [1,3,1,2,2,3 - 12 marks]

Isagani's team 55, 67, 44, 48, 57, 56, 88, 101, 95, 34, 65, 63, 75, 81, 73

Lara's team 77, 74, 98, 100, 54, 61, 83, 92, 87, 78, 65, 57, 85, 81, 73

a) Is the above data continuous or discrete?

b) Display the above data using the appropriate graphical display



c) Describe the shape of both team's data.

Isagani's Positive skew
Lara Symmetrical.

d) Calculate each team's:

i) Mean

Isagani's 66.8
Lara - 77.7

ii) Median

Isagani's 65
Lara 78

iii) Decide which team performed better last year? Justify your answer.

Lara reason

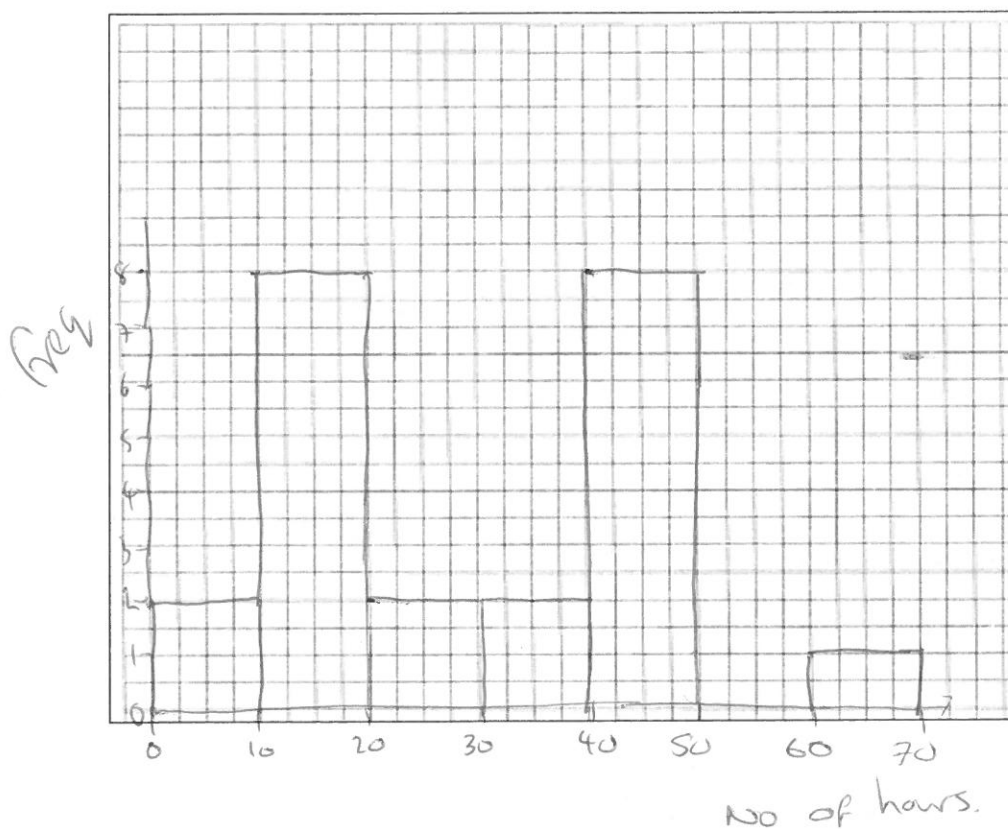
3) The data below is information about the number of hours worked by different employees during 1 week at Built Fast Enterprises. [3 + 4 + 2 + 3 = 12 marks]

Week 1 data: 4, 8, 11, 12, 13, 13, 15, 17, 18, 19, 29, 29, 33, 35, 40, 41, 42, 42, 43, 45, 48, 49, 62

a. create a frequency table for this data.

0 - <10	2
10 - <20	8
20 - <30	2
30 - <40	2
40 - <50	8
50 - <60	0
60 - <70	1

b. Use the frequency table to create a histogram.



c. Comment on the shape of the data in the histogram.

bimodal.

- d. Later, Built Fast analysed two separate weeks to look at the productivity of their employees during their working hours. They compiled their results below:



Comment on the distribution of both data sets and explain which week was more productive, using specific reference to how each week's distribution impacts the measures of centre.

Week 1 is a negative distribution. Week 2 is a positive distribution.

Link - Because of wk 1 negative distribution, the mean is less than the median and mode.

Link - Because of wk 2 positive distribution, the mean is greater than the median and mode.

Link - Any other reasonable reference to why week 1 was more productive.

Link - week 1 was more productive.

2 mks - Any 2 reasonable justifications, given they aren't repetitive.